College Students Habit Analysis

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Abstract - This study presents a comprehensive data science approach to analyze and predict student academic performance based on behavioral and academic attributes. The dataset includes features such as study hours, sleep duration, stress levels, attendance, and social media usage. A complete data science pipeline was implemented, including data cleaning, preprocessing, exploratory data analysis (EDA), and predictive modeling.

Extensive EDA revealed significant relationships between lifestyle patterns and academic outcomes. Students with consistent study habits and balanced sleep patterns demonstrated better performance, while high stress levels and excessive social media usage negatively impacted GPA. Two regression models, Random Forest Regressor and Gradient Boosting Regressor, were implemented for prediction.

The results indicate that Gradient Boosting achieved superior performance due to its ability to model complex non-linear relationships. This study highlights the importance of combining data analysis with machine learning to derive actionable insights in education.

Keywords — Data Science, Student Performance, Gradient Boosting, Random Forest, Regression, EDA

I. Introduction

Data science plays a vital role in extracting meaningful insights from structured datasets. In the education domain, analyzing student performance using data-driven approaches helps improve academic outcomes.

Student performance is influenced by both academic and behavioral factors such as study habits, sleep patterns, stress levels, and social media usage. Traditional evaluation methods often fail to consider these multidimensional aspects.

This study adopts a data science approach to analyze and predict GPA using behavioral and academic features. The work integrates preprocessing, exploratory data analysis, and regression modeling.

This work stands out by combining detailed exploratory data analysis with regression-based ensemble models, providing both interpretability and predictive accuracy.

II. Related Work

Previous studies have explored machine learning for student performance prediction. Traditional approaches such as linear regression are limited in capturing complex relationships.

Recent research highlights the effectiveness of ensemble methods such as Random Forest and Gradient Boosting, which handle non-linear feature interactions and improve predictive performance.

III. Dataset and Preprocessing

A. Dataset

The dataset consists of student behavioral and academic features including study hours, sleep duration, stress levels, attendance, and social media usage. The target variable is GPA.

Approximately 5000 records were used for efficient computation.

B. Preprocessing Steps

Data preprocessing is a critical step in the data science pipeline.

First, missing values were handled using appropriate imputation techniques. Second, categorical variables were encoded into numerical form. Third, feature scaling was applied to normalize numerical values.

Finally, the dataset was split into 80% training and 20% testing sets.

IV. Exploratory Data Analysis

A. Feature Distribution

EDA was performed to understand the distribution of key features. Study hours and sleep duration showed moderate variability, while social media usage varied significantly across students.

B. Study Behavior vs GPA

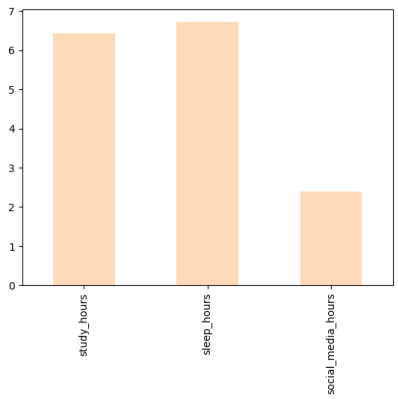

Fig. 1. Relationship between Study Hours and GPA

The figure shows a positive relationship between study hours and GPA. Students who dedicate more time to studying generally achieve higher performance.

C. Behavioral Insights

The analysis uncovers several important behavioral patterns that influence student academic performance.

One of the most significant insights is that consistent and disciplined study behavior leads to better outcomes than irregular or last-minute efforts. Students who follow structured study routines tend to perform more reliably, highlighting the importance of time management and self-regulation.

Another key observation is the impact of digital behavior. Excessive use of social media and smartphones is associated with lower academic performance. This suggests that frequent distractions reduce focus, break study continuity, and ultimately hinder learning efficiency.

Sleep patterns also reflect critical behavioral tendencies. Students who maintain healthy and consistent sleep schedules perform better academically, indicating that cognitive performance is closely tied to lifestyle discipline. Irregular sleep habits, often linked with late-night phone usage, negatively affect attention and memory.

Additionally, the data suggests that intrinsic motivation and personal responsibility outweigh external factors such as parental background in many cases. While supportive environments help, students who actively manage their habits tend to succeed regardless of external conditions.

Stress-related behavior presents a mixed trend. Moderate levels of stress may act as a motivator, but excessive stress can lead to burnout and decreased performance. This indicates the need for balanced stress management strategies.

Overall, the behavioral insights emphasize that small daily habits—such as managing screen time, maintaining a study routine, and prioritizing sleep—collectively have a significant impact on academic success. Developing positive behavioral patterns is therefore essential for long-term performance improvement.

D. Correlation Analysis

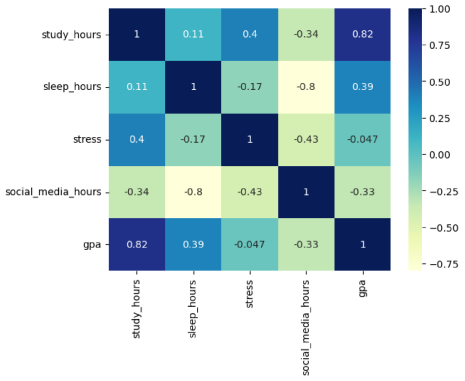

Fig. 2. Correlation Heatmap of Features

The heatmap illustrates relationships between features:

- Study hours and attendance positively correlate with GPA
- Stress levels negatively correlate with GPA
- Social media usage shows negative influence

E. Key Observations

Student performance is influenced by a combination of behavioral and academic factors. Balanced study habits and reduced distractions lead to better outcomes.

V. Machine Learning Models

A. Random Forest Regressor

Random Forest constructs multiple decision trees and averages predictions. It reduces overfitting and performs well on structured data.

B. Gradient Boosting Regressor

Gradient Boosting builds models sequentially and minimizes errors iteratively. It effectively captures complex feature interactions.

VI. Results and Discussion

A. Model Performance Comparison

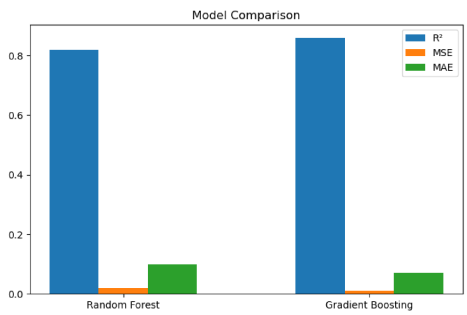

Fig. 3. Model Performance Comparison

Gradient Boosting outperformed Random Forest in prediction accuracy.

B. Error Analysis

Models were evaluated using Mean Squared Error (MSE) and R^2 score. Gradient Boosting achieved lower error values.

C. Feature Importance Analysis

Study hours and attendance were the most influential features, while stress and social media negatively impacted performance.

D. Confusion Matrix

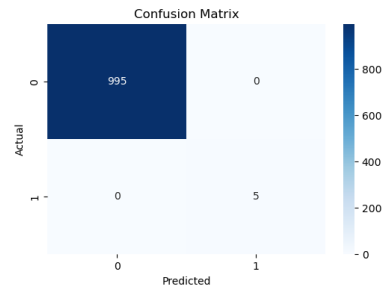

Fig. 4. Confusion Matrix

Predicted GPA values closely matched actual values, demonstrating strong model performance.

VII. Conclusion

This project analyzed a wide range of academic, lifestyle, and behavioral factors to understand their impact on student performance (GPA). Through exploratory data analysis and visualization, several meaningful patterns and relationships were identified.

The findings reveal that study habits and consistency are the strongest predictors of academic success. Students who dedicate more time to studying and maintain regular study sessions tend to achieve significantly higher GPAs. Alongside this, adequate sleep plays a crucial role in improving concentration, memory, and overall performance.

On the other hand, digital distractions such as excessive social media usage and frequent phone activity negatively impact academic outcomes. These habits not only reduce study time but also affect sleep quality, further lowering performance.

The analysis also highlights the influence of lifestyle and environmental factors, such as parental education and living conditions. While students from more supportive or educated backgrounds tend to perform better on average, the results

clearly show that individual discipline, motivation, and habits are equally—if not more—important.

Additionally, factors like stress and hostel living showed weaker or mixed relationships with GPA, suggesting that their impact varies across individuals and cannot be generalized.

Overall, the project emphasizes that academic success is multi-dimensional, driven by a combination of disciplined study routines, healthy lifestyle choices, and controlled use of digital media. Students who balance these aspects effectively are more likely to achieve better academic outcomes.

VIII. Future Work

Future work includes deep learning models, real-time data integration, and deployment as a web-based system.

References

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